

# Md Shahid Ali Khan

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## PROFILE SUMMARY

Backend Software Engineer with **3+ years of experience** in **Java and Spring Boot**, building **microservices and distributed systems** using **Kafka, Redis, REST APIs, concurrency, and asynchronous processing** in the **financial services domain**.

## TECHNICAL SKILLS

**Languages & Core CS:** Java, Data Structures & Algorithms, OOP, Operating Systems, DBMS

**Backend & Distributed Systems:** Spring Boot, Spring Data JPA, Hibernate, REST APIs, Microservices, Spring WebFlux, Kafka, Redis, Caching, Concurrency, Multithreading, CompletableFuture, Resilience4j, System Design

**Databases, Infrastructure & Tools:** MySQL, PostgreSQL, Docker, Kubernetes, Jenkins (CI/CD), Git

## WORK EXPERIENCE

### Societe Generale – Software Engineer

July 2023 – Present

Report Generation & Risk Analytics Platform | *Spring Boot, Microservices, Kafka, Redis*

Bangalore

- **Drove the development and modernization** of a financial risk analytics platform generating **NPV, GAP, and NIM reports**, evolving it from a **modular monolith** toward independently deployable microservices.
- Designed a **two-level caching architecture** using an in-memory **L1 cache** and **Redis-based L2 distributed cache**, reducing API response time from **10 seconds to under 2 seconds** for optimized report requests.
- Optimized **cache keys, TTLs, and eviction**, cutting report generation time from **2 minutes to 30 seconds**.
- Refactored long-running report generation into **Kafka-driven asynchronous workflows**, enabling **concurrent report processing** and improving API responsiveness.
- Implemented resilient **Kafka consumer workflows** with **retries, Dead Letter Topics (DLTs), and automated email notifications** for failed report batches.
- Processed **50K+ records per report execution** across financial risk analysis and reporting workloads.

### Societe Generale – Software Developer Intern

March 2023 – July 2023

*Rejected Transaction Impact Analysis & Script Migration*

Bangalore

- Built a **Spring Boot**-based analysis platform to categorize rejected transactions by risk, processing **100K+ transaction records** using optimized SQL queries to accelerate remediation workflows.
- Migrated **70+ legacy Unix shell scripts** to Linux and refactored workflows, improving system compatibility, maintainability, and operational efficiency.

## PROJECTS

Distributed Workflow & Job Processing Platform | *Java 25, Spring Boot, Redis, Kafka, PostgreSQL, Docker, Kubernetes, Lua*

- Built a **fault-tolerant distributed workflow platform** on a local **3-node Kubernetes (Kind) cluster**, combining **Redis-backed rate limiting, caching, distributed locking, leader election, and Kafka-based asynchronous job processing**.
- Implemented **Redis/Lua-based atomic rate limiting**, sustaining **4K+ RPS** with **70ms P95 / 120ms P99 latency** at 5x normal load; achieved a **90%+ cache hit ratio**, reducing read latency from **50ms to under 5ms** on cache hits.
- Built a **Kafka-based job processing pipeline** with **at-least-once delivery, idempotent consumers, exponential backoff, and DLQ recovery**, processing **10,000+ jobs** across 3–5 workers without duplicate side effects during forced retries.
- Implemented **lease-based distributed locks with fencing tokens** to prevent stale-lock writes and **leader election** for cluster-wide coordination, achieving **5–10 second failover** after leader pod termination.
- Performed **resilience and failure-injection testing** by simulating pod, Redis, and Kafka failures, validating **job recovery, lock handoff, leader re-election, and rate-limit/cache consistency**.
- Added **Zipkin distributed tracing** across the rate-limit, cache, job-queue, locking, and leader-election workflows to identify **latency bottlenecks and failure points**.

## EDUCATION

### BIT Mesra

Ranchi, India

*B.Tech in Information Technology | GPA: 8.60/10*

2019 – 2023

- Received **“Spot” and “Newbie” Awards** at Societe Generale for strong performance and contributions.
- Secured **National Rank 844** in CODEKAZE 2021 among **200,000+ participants**.
- Scored in the **top 1 percentile** in JEE Main among **1 million+ candidates**.
- Awarded the merit-based **GP Birla Scholarship** at BIT Mesra, covering **50% of tuition fees**.